

**ZAHEDAN INTL, Iran, Islamic Republic of**WMO: **408560**Lat: **29.4720N**Lon: **60.9006E**Elev: **4495**StdP: **12.46**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 23.0       | 26.5 | -11.3                         | 3.5 | 44.9 | -4.4 | 5.1 | 47.9 | 29.9                  | 59.1 | 27.3 | 60.4 | 3.2                   | 170  | 0.489 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 27.5                   | 102.5           | 61.8 | 100.6 | 61.1 | 98.7 | 60.1 | 65.3                | 94.4 | 63.3 | 95.4 | 61.9 | 94.5 | 10.6                 | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 54.0                            | 73.4 | 73.8 | 50.9 | 65.3 | 69.5 | 48.3 | 59.2 | 69.0 | 32.8          | 93.4 | 31.1 | 96.0 | 30.0 | 94.2 | 75.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 25.4          | 21.8 | 18.6 | 16.5                       | 107.3 | 4.1                | 3.1 | 13.5                                               | 109.5 | 11.1       | 111.4 | 8.8        | 113.1 | 5.8        | 115.4 |    |    |
| (5)               |      |      | 12.6                       | 67.8  | 4.2                | 3.2 | 9.6                                                | 70.1  | 7.1        | 72.0  | 4.7        | 73.8  | 1.7        | 76.1  |    |    |

**Monthly Climatic Design Conditions**

|             |                                                                                  | Annual  | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|-------------|----------------------------------------------------------------------------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|             |                                                                                  | (d)     | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)         | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg   | 66.7  | 46.2  | 51.0  | 60.2  | 69.7  | 78.0  | 83.0  | 84.9  | 81.0  | 74.7  | 67.0  | 55.6  | 47.8 | (6)  |
| (7)         |                                                                                  | DBStd   | 14.92 | 7.96  | 8.78  | 7.76  | 6.72  | 5.25  | 4.31  | 3.66  | 3.80  | 4.65  | 5.48  | 7.51  | 7.28 | (7)  |
| (8)         |                                                                                  | HDD50   | 427   | 171   | 86    | 12    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 29    | 129  | (8)  |
| (9)         |                                                                                  | HDD65   | 2071  | 583   | 395   | 187   | 33    | 1     | 0     | 0     | 3     | 41    | 293   | 535   | (9)  |      |
| (10)        |                                                                                  | CDD50   | 6510  | 53    | 115   | 329   | 591   | 867   | 991   | 1080  | 960   | 740   | 528   | 197   | 59   | (10) |
| (11)        |                                                                                  | CDD65   | 2676  | 0     | 3     | 39    | 173   | 403   | 541   | 615   | 495   | 292   | 104   | 11    | 0    | (11) |
| (12)        |                                                                                  | CDH74   | 41274 | 5     | 37    | 459   | 2196  | 5528  | 8442  | 10055 | 7952  | 4549  | 1835  | 203   | 13   | (12) |
| (13)        |                                                                                  | CDH80   | 23323 | 0     | 3     | 96    | 860   | 2957  | 5161  | 6398  | 4768  | 2386  | 668   | 26    | 0    | (13) |
| (14)        | Wind                                                                             | WSAvg   | 7.8   | 8.3   | 9.8   | 9.6   | 8.5   | 7.9   | 8.6   | 8.4   | 7.4   | 6.5   | 5.7   | 6.4   | 6.8  | (14) |
| (15)        | Precipitation                                                                    | PrecAvg | 3.3   | 0.9   | 0.5   | 0.5   | 0.4   | 0.1   | 0.1   | 0.0   | 0.0   | 0.1   | 0.2   | 0.6   | (15) |      |
| (16)        |                                                                                  | PrecMax | 8.5   | 4.3   | 2.5   | 2.3   | 1.6   | 0.7   | 0.9   | 0.8   | 0.2   | 0.1   | 1.3   | 1.1   | 4.0  | (16) |
| (17)        |                                                                                  | PrecMin | 0.6   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  | (17) |
| (18)        |                                                                                  | PrecStd | 1.9   | 0.9   | 0.7   | 0.5   | 0.4   | 0.2   | 0.2   | 0.2   | 0.0   | 0.0   | 0.3   | 0.2   | 0.9  | (18) |
| (19)        | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%    | DB    | 73.4  | 78.9  | 87.5  | 94.6  | 101.2 | 105.2 | 105.5 | 102.6 | 98.3  | 93.2  | 82.7  | 75.5 | (19) |
| MCWB        |                                                                                  |         | 48.6  | 51.7  | 53.2  | 58.4  | 61.7  | 61.5  | 62.2  | 61.5  | 59.3  | 57.5  | 53.0  | 50.6  | (20) |      |
| 2%          |                                                                                  | DB      | 68.2  | 73.6  | 82.4  | 91.2  | 98.4  | 102.3 | 102.9 | 100.3 | 95.6  | 89.7  | 79.3  | 70.2  | (21) |      |
|             |                                                                                  | MCWB    | 47.8  | 50.6  | 52.3  | 56.8  | 59.9  | 61.2  | 62.2  | 60.4  | 57.9  | 56.9  | 51.9  | 48.2  | (22) |      |
| 5%          |                                                                                  | DB      | 64.7  | 69.9  | 79.0  | 88.1  | 95.3  | 100.2 | 100.8 | 98.2  | 93.6  | 86.4  | 75.9  | 66.5  | (23) |      |
|             |                                                                                  | MCWB    | 45.8  | 49.1  | 51.6  | 55.6  | 58.8  | 60.6  | 61.6  | 59.4  | 57.2  | 55.7  | 51.3  | 46.7  | (24) |      |
| 10%         |                                                                                  | DB      | 61.2  | 66.3  | 75.4  | 85.6  | 93.0  | 98.2  | 98.9  | 96.4  | 91.5  | 83.9  | 71.9  | 63.1  | (25) |      |
|             |                                                                                  | MCWB    | 45.0  | 47.7  | 50.8  | 55.0  | 58.2  | 59.6  | 60.7  | 58.8  | 56.4  | 54.3  | 49.9  | 45.2  | (26) |      |
| (27)        | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%    | WB    | 52.6  | 54.7  | 57.5  | 60.0  | 64.0  | 66.3  | 70.2  | 67.8  | 62.1  | 61.4  | 56.5  | 52.8 | (27) |
| MCD B       |                                                                                  |         | 63.8  | 69.3  | 74.4  | 87.6  | 93.7  | 89.3  | 88.8  | 87.8  | 91.8  | 84.1  | 72.4  | 67.5  | (28) |      |
| 2%          |                                                                                  | WB      | 50.4  | 52.7  | 55.6  | 58.4  | 61.7  | 63.7  | 66.2  | 63.3  | 60.2  | 58.3  | 54.6  | 50.6  | (29) |      |
|             |                                                                                  | MCD B   | 62.8  | 67.2  | 72.6  | 84.5  | 92.0  | 95.7  | 93.7  | 95.2  | 91.3  | 83.9  | 72.7  | 65.3  | (30) |      |
| 5%          |                                                                                  | WB      | 48.8  | 51.1  | 54.1  | 57.3  | 60.3  | 62.1  | 64.0  | 61.4  | 58.7  | 56.8  | 53.1  | 49.1  | (31) |      |
|             |                                                                                  | MCD B   | 61.3  | 65.3  | 70.8  | 82.3  | 90.6  | 95.1  | 96.6  | 94.4  | 90.1  | 82.8  | 71.4  | 63.3  | (32) |      |
| 10%         |                                                                                  | WB      | 46.7  | 49.8  | 52.8  | 56.2  | 59.3  | 60.8  | 62.4  | 59.9  | 57.3  | 55.4  | 51.3  | 47.2  | (33) |      |
|             |                                                                                  | MCD B   | 59.1  | 63.5  | 69.9  | 80.2  | 89.0  | 94.7  | 95.5  | 93.3  | 88.9  | 81.0  | 69.1  | 60.5  | (34) |      |
| (35)        | Mean Daily<br>Temperature<br>Range                                               | 5% DB   | MDBR  | 24.6  | 24.2  | 24.1  | 25.6  | 26.0  | 27.2  | 27.5  | 30.0  | 32.3  | 31.1  | 28.6  | 27.5 | (35) |
| MCDBR       |                                                                                  |         | 27.5  | 25.1  | 27.6  | 29.2  | 29.8  | 30.4  | 30.1  | 31.4  | 34.0  | 33.2  | 31.3  | 30.5  | (36) |      |
| MCWBR       |                                                                                  |         | 14.9  | 11.9  | 11.7  | 11.6  | 11.9  | 11.7  | 11.7  | 12.4  | 14.7  | 15.4  | 15.1  | 16.4  | (37) |      |
| 5% WB       |                                                                                  | MCDBR   | 20.6  | 19.7  | 21.2  | 24.7  | 26.8  | 26.7  | 25.9  | 27.4  | 31.0  | 30.0  | 26.2  | 24.4  | (38) |      |
|             |                                                                                  | MCWBR   | 12.2  | 10.0  | 9.5   | 10.1  | 10.7  | 10.9  | 10.6  | 11.4  | 13.8  | 14.2  | 13.2  | 13.6  | (39) |      |
|             |                                                                                  |         |       |       |       |       |       |       |       |       |       |       |       |       |      | (39) |
| (40)        | Clear-Sky<br>Solar<br>Irradiance                                                 | taub    | 0.319 | 0.370 | 0.422 | 0.486 | 0.505 | 0.481 | 0.465 | 0.429 | 0.393 | 0.390 | 0.339 | 0.308 | (40) |      |
| taud        |                                                                                  | 2.388   | 2.202 | 2.019 | 1.882 | 1.850 | 1.908 | 1.961 | 2.049 | 2.131 | 2.131 | 2.316 | 2.416 | (41)  |      |      |
| Ebn at Noon |                                                                                  | 291     | 282   | 273   | 259   | 254   | 259   | 263   | 272   | 279   | 271   | 279   | 289   | (42)  |      |      |
| Edh at Noon |                                                                                  | 32      | 42    | 54    | 64    | 66    | 62    | 59    | 53    | 47    | 44    | 34    | 30    | (43)  |      |      |
| (44)        | All-Sky Solar<br>Radiation                                                       | RadAvg  | 1211  | 1437  | 1797  | 2116  | 2335  | 2530  | 2449  | 2365  | 2115  | 1721  | 1352  | 1165  | (44) |      |
| (45)        |                                                                                  | RadStd  | 80    | 94    | 103   | 93    | 82    | 90    | 83    | 87    | 48    | 49    | 53    | 70    | (45) |      |

**Historical Trends**

|      |                        | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page